Study Protocol

1. Study Information

Study title	MR-Driven Virtual Fixture Generation and Physical Human-Robot Interaction Study
Principal investigator	
Institution	
Study site	
Researcher contact	
Ethics approval	
Estimated duration	Approximately 120 minutes

This document describes what you will do during the study session. Please also read the Participant Information and Informed Consent Form for information about your rights, data protection, risks, and withdrawal.

2. Purpose of the Study

This study investigates different ways of creating and using motion guidance for collaborative robots. It compares several path-authoring approaches and several forms of robot guidance during a physical human-robot task. We are interested in task performance, workload, and usability.

3. Who Can Take Part?

You may take part if you meet the study eligibility requirements. In general, participants must:

- Be an adult who is able to provide informed consent.
- Have normal or corrected-to-normal vision sufficient to view the task environment and use the mixed-reality headset.
- Be able to use the study interfaces, including handheld controls, mouse and keyboard, and physical robot interaction.

You should not take part if any of the following applies:

- You have a condition that would make mixed-reality headset use or physical interaction with a collaborative robot unsafe.
- You have severe susceptibility to motion sickness or have previously been unable to tolerate immersive headsets.
- There is another reason identified by you or the researcher that would make participation unsafe.

4. Equipment Used in the Study

Equipment	**How it is used**
Flexiv Rizon 4 collaborative robot	Used for the physical steady-hand task.
Ring-shaped robot end-effector	The ring-shaped tool that you guide along the spatial wire.
Spatial wire apparatus	The three-dimensional reference path used for the steady-hand task.
Meta Quest 3 headset	Used for mixed-reality path generation and path editing.
Screen-based Unity interface	Used to specify key robot poses with mouse-and-keyboard input.
OptiTrack tracking system	Used to demonstrate a desired six-degree-of-freedom motion path.

5. Experimental Task

You will perform a collaborative steady-hand (buzz-wire) task. A ring-shaped end-effector mounted on the robot must be guided along a spatial wire while avoiding contact. The task requires you to control both the position and orientation of the robot end-effector.

6. Study Conditions

6.1 Path-Authoring Approaches

Label	Method	What you will do
SBS	Screen-Based Specification	Place and orient key poses using the screen-based Unity interface and mouse/keyboard.
OTD	OptiTrack-Based Demonstration	Demonstrate the desired motion using a handheld tracked object.
MRG	MR-Driven Generation	Use the Meta Quest 3 interface to move a virtual proxy and generate a robot path.
MRGE	MR-Driven Generation with Editing	Generate a path in mixed reality and then adjust selected positions and/or orientations using virtual editing handles.

6.2 Robot-Guidance Conditions

Label	Condition	Guidance during the task
NO-VF	No virtual fixture	You guide the robot without virtual-fixture assistance.
SBS	Guidance from the SBS path	Translational and rotational guidance based on the SBS-authored path.
OTD	Guidance from the OTD path	Translational and rotational guidance based on the OTD-authored path.
MRG	Guidance from the MRG path	Translational and rotational guidance based on the MRG-authored path.
MRGE-T	Translational guidance only	Uses the MRGE-authored path with translational guidance but without rotational guidance.
MRGE	Full MRGE guidance	Translational and rotational guidance based on the MRGE-authored path.

7. What Will Happen During the Session?

7.1 Check-in, Safety Briefing, and Consent

The researcher will explain the study, equipment, safety procedures, and emergency stop. You will have time to ask questions. Written informed consent will be obtained before any experimental task, and you will be assigned a participant ID for study records.

7.2 Familiarization and Practice

Before formal evaluation trials begin, you will receive approximately 90 minutes of hands-on familiarization and practice. Practice data collected during this training period will not be included in the study analysis.

- The researcher will first demonstrate the steady-hand task and safe physical interaction with the robot.
- You will complete three practice traversals of the steady-hand task without virtual-fixture guidance.
- For each of the four path-authoring approaches (SBS, OTD, MRG, and MRGE), the researcher will demonstrate the interaction and you will complete three practice trials.
- For each practice path, you will complete three practice traversals with virtual-fixture guidance so that you can become familiar with the assisted robot motion before the formal evaluation trials.

- After the standard practice, you will be given additional practice time and may choose which interfaces or interactions you would like to practice further before the formal evaluation trials begin. The researcher can provide assistance as needed.

7.3 Path-Authoring Evaluation

You will complete one formal evaluation trial with each of the four path-authoring approaches: SBS, OTD, MRG, and MRGE.

7.4 Task-Execution Evaluation

You will complete one formal steady-hand evaluation trial under each of the six robot-guidance conditions: NO-VF, SBS, OTD, MRG, MRGE-T, and MRGE.

7.5 Questionnaires and Feedback

For selected conditions (NO-VF, SBS, OTD, and MRGE), you will complete the NASA Task Load Index (NASA-TLX) and the System Usability Scale (SUS). These questionnaires ask about your workload and your experience using the study system. You may also provide brief written comments about usability, difficulties, or suggested improvements.

7.6 Breaks and Completion

You may request a break at any time. At the end of the session, the researcher will check that you feel well, answer any final questions, and remind you how to contact the research team if you have questions after the study.

8. What Information Will Be Recorded?

The study will record task-related information such as the paths you create, robot motion and interaction data, task completion time, path-following performance, contact with the task wire, questionnaire responses, and optional written comments. No audio, video, photographs, or screen recordings are planned for this study.

9. Risks, Breaks, and Stopping the Study

Possible temporary discomforts include eye strain, dizziness, nausea, fatigue, frustration, or minor accidental contact with the robot or task apparatus.

Possible issue	**What will be done**
Headset discomfort	You may remove the headset immediately. The MR portion will be paused or stopped if symptoms continue.
Physical fatigue or pain	You may rest at any time. The task will stop if you report pain or significant fatigue.
Unexpected robot or tracking behavior	The researcher will stop the trial immediately if unsafe motion, tracking failure, or another technical problem occurs.
Frustration or performance pressure	You may ask questions, take a break, skip optional questionnaire items, or stop participating without penalty.

You may pause or stop at any time. The researcher will also stop a trial if there is a safety concern, and it will not resume until both you and the researcher are satisfied that it is safe to continue.

If you feel unwell, are injured, or have any concern, tell the researcher immediately. Relevant procedures are described in the Participant Information and Informed Consent Form.